Quantification: 682_z23
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	12
interior lacuna count	11
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2088.2	1801	1031.95	px^2
major_axis_length_px	lacuna shape	85.9334	87.8115	23.7232	px
minor_axis_length_px	lacuna shape	31.5731	30.3414	8.451	px
aspect_ratio	lacuna shape	2.8283	2.7498	0.7793	unitless
eccentricity	lacuna shape	0.9147	0.9291	0.0652	unitless
solidity	lacuna shape	0.8718	0.8885	0.0596	unitless
roots_count	headline	10	10	3.2249	count
ring_length_r30_px	headline	393.818	421	103.59	px
ring_length_r60_px	reported	950.091	980	196.177	px
owned_length_px	ownership-dependent	3114.13	2519	1787.96	px
edge_count	ownership-dependent	148.182	102	91.7658	count
mean_edge_length_px	ownership-dependent	21.4187	20.9043	2.0973	px
perimeter_px	normalised	207.522	210.167	71.7085	px
ring_area_r30_px2	normalised	8536.09	8646	1979.1	px^2
ring_area_r60_px2	normalised	21938.7	22056	3981.31	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9972	1	0.0094	unitless
ring_density_r30	normalised	0.0466691	0.0486364	0.00927396	px^-1
ring_density_r60	normalised	0.0438535	0.0452485	0.00808305	px^-1
roots_per_100px_perimeter	normalised	5.364	4.7701	2.6171	per 100 px
ring_attached_length_r30_px	network v2	360	384	99.7918	px
ring_attached_length_r60_px	network v2	781.636	825	205.059	px
ring_length_w_r30_px	network v2	445.793	493.446	115.818	px
ring_length_w_r60_px	network v2	1075.18	1099.67	216.988	px
sholl_crossings_r10	network v2	9.1818	10	3.4876	count
sholl_crossings_r20	network v2	11.9091	12	3.145	count
sholl_crossings_r30	network v2	13.1818	13	3.7368	count
loop_count	loops, report only	14.9091	8	13.4941	count
loop_edge_share	loops, report only	0.3401	0.3286	0.1253	share
ring_width_mean_r30_px	width	6.8526	6.8027	0.528	px

Per field
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measure	value	units
analysed_area_px2	1.0246e+06	px^2
total_skeleton_length_px	38979	px
canalicular_length_density_per_px	0.0380433	px^-1
junction_count	928	count
junction_density_per_px2	0.00090572	px^-2
lacuna_count	12	count

lacunae_per_px2	1.171e-05	px^-2
skeleton_component_count	246	count
mean_component_length_px	158.451	px
median_component_length_px	16.5	px
field_length_density_w_per_px	0.0431096	px^-1
field_density_without_flagged_per_px	0.0390762	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	560	px
unattached_component_count	30	count
canalicular_length_density_attached_per_px	0.0380433	px^-1
field_length_density_with_unattached_per_px	0.0385899	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	196	count
loop_edge_share	0.3773	share
edge_count	2025	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.806	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10.198	px
width_pixels_used	37539	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1379	89.4664	21.3431	4.1918	0.9711	0.7971	-1.5436	60.8564	471.593	False
2	4132	136.42	39.4244	3.4603	0.9573	0.9008	1.5344	431.979	214.138	False
3	1032	65.9485	21.7201	3.0363	0.9442	0.8418	-0.9507	441.636	703.582	False
4	2868	91.0398	40.6809	2.2379	0.8946	0.9166	-1.49	534.719	477.113	False
5	980	60.8458	24.7024	2.4632	0.9139	0.7644	-1.4596	557.819	348.526	False
6	2962	111.965	35.5084	3.1532	0.9484	0.917	1.4275	591.514	47.6516	True
7	2810	108.692	33.8251	3.2134	0.9503	0.9201	1.4475	714.117	172.09	False
8	1297	66.0375	26.8577	2.4588	0.9136	0.9134	-1.189	725.743	851.213	False
9	1519	86.1566	25.1812	3.4215	0.9563	0.8388	1.3918	837.228	113.164	False
10	2083	63.4325	42.3049	1.4994	0.7451	0.9485	-1.2565	905.171	752.206	False
11	2782	91.2948	39.6911	2.3001	0.9005	0.8762	-1.4405	919.61	300.233	False
12								180.095	840.839	False

Network per lacuna

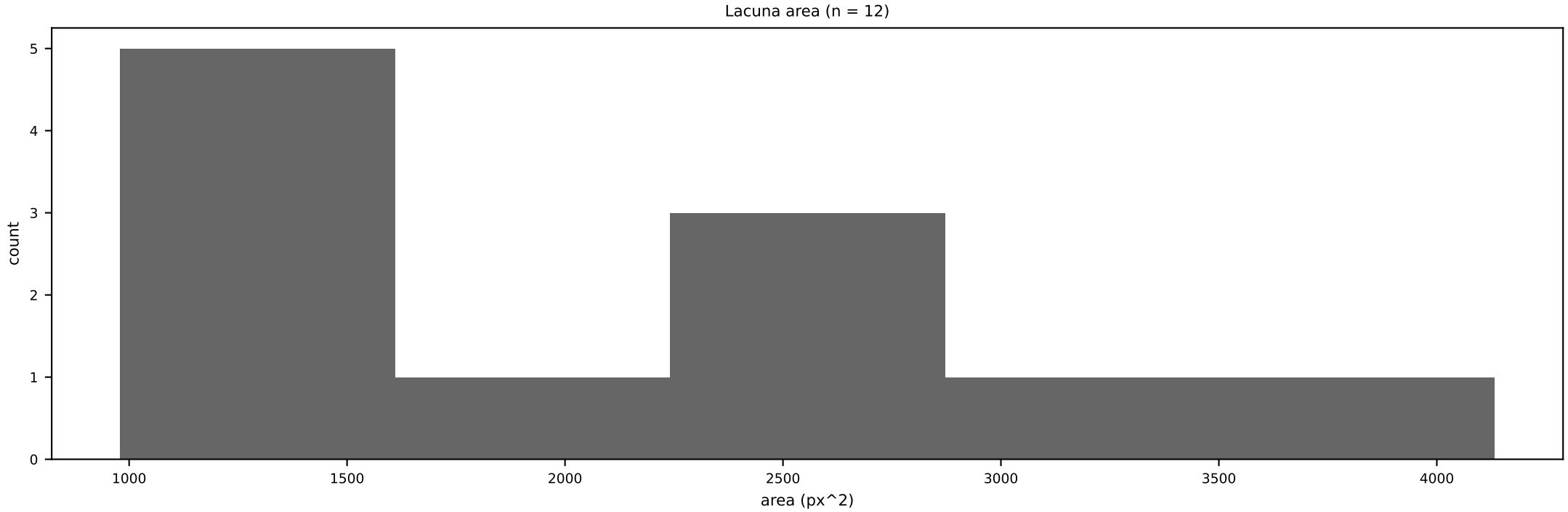
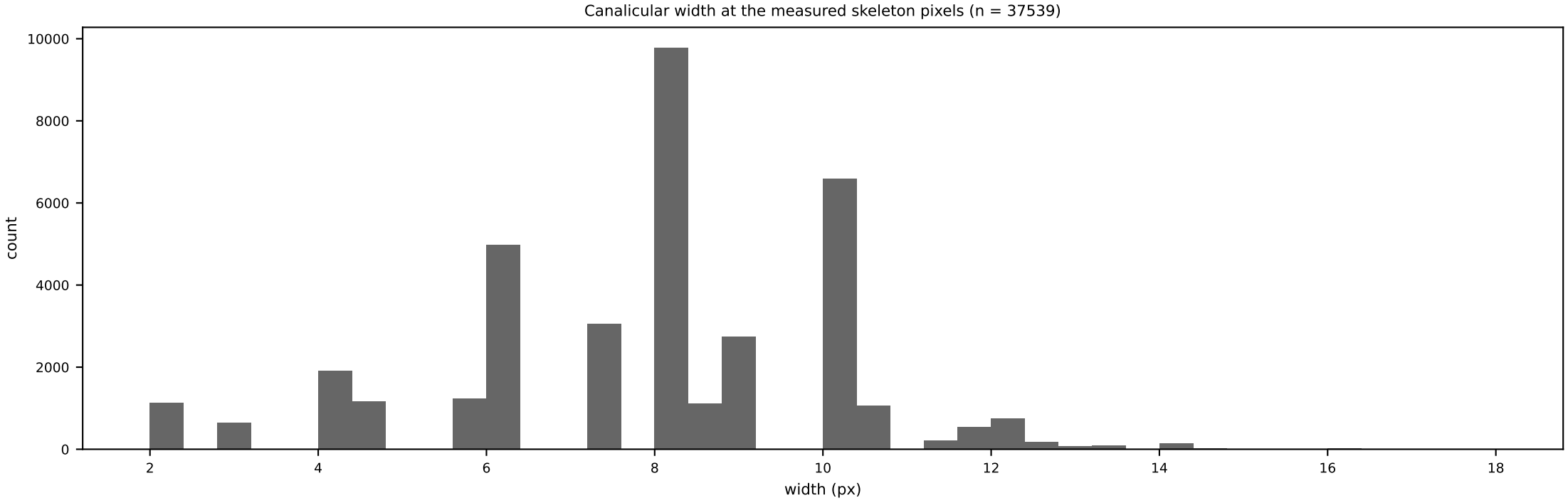
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	8	386	840	1827.5	92	19.8641	6.3283
2	16	511	1163	6179.55	304	20.3275	7.5743
3	5	421	1046	4046.57	198	20.4372	7.6867
4	13	465	998	4484.45	212	21.153	6.8683
5	10	460	980	1421.29	74	19.2066	7.4533

6	10	351	815	1097.88	52	21.1131	7.4606
7	10	537	1356	2519	102	24.696	6.9511
8	8	321	749	2786.73	108	25.8031	6.6161
9	14	442	1011	1529.94	69	22.173	6.0426
10	9	276	809	1507.2	70	21.5314	6.8027
11	10	299	803	1944.1	93	20.9043	6.5377
12	7	214	696	6009.08	308	19.51	6.5174

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	219.681	8976	22806	1	0.9689	0.0430036	0.0368324	3.6416
2	335.421	12178	29932	1	1	0.0419609	0.0388547	4.7701
3	167.238	7485	20582	1	1	0.0562458	0.0508211	2.9898
4	243.522	9472	22056	1	1	0.0490921	0.0452485	5.3383
5	167.924	7349	17555	1	1	0.0625935	0.0558246	5.9551
6	282.894	8678	19332	0.813	0.7462	0.0404471	0.0421581	3.5349
7	263.48	10172	24552	1	1	0.052792	0.0552297	3.7954
8	174.51	7570	20770	1	1	0.0424042	0.0360616	4.5843
9	210.167	8646	22189	1	1	0.0511219	0.0455631	6.6614
10	185.823	7980	21610	1	1	0.0345865	0.0374364	4.8433
11	259.279	9669	24825	1	1	0.0309236	0.0323464	3.8568
12	55.6985	4400	14449	1	1	0.0486364	0.0481694	12.5677

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	300	526	428.463	928.068	10	12	14
2	482	916	566.475	1295.76	10	16	14
3	399	841	493.446	1207.89	11	13	13
4	439	828	524.475	1128.31	12	14	12
5	448	951	520.162	1099.67	11	13	15
6	259	547	400.191	919.21	9	12	11
7	490	1219	599.061	1524.3	15	16	21
8	277	637	362.321	864.051	6	12	10
9	384	825	514.316	1148.35	9	12	17
10	266	664	315.836	932.749	9	9	11
11	272	543	336.693	899.926	6	8	11
12	203	648	242.48	797.926	2	6	7

682_z23: distributions, pre-validation, px